

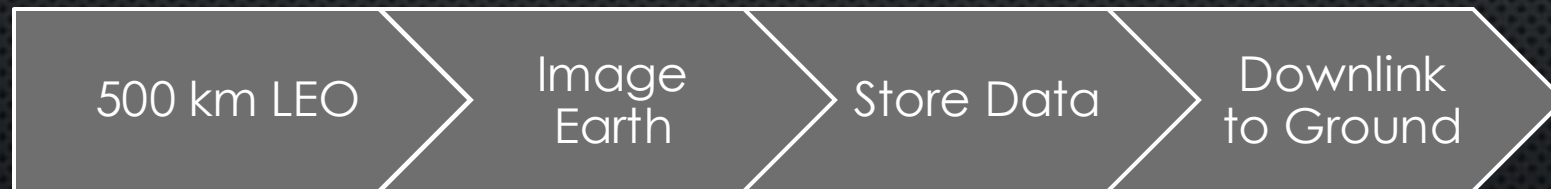


AURORA-1 | 2U EARTH- OBSERVATION CUBESAT

BY CHANCELLOR DELK

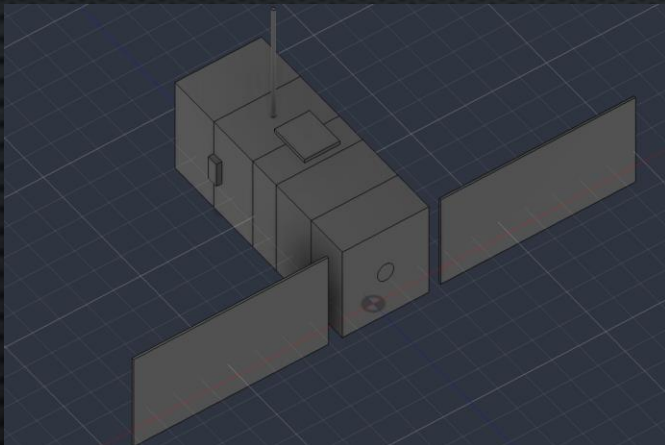
MISSION OBJECTIVE

- DESIGN A DIGITAL 2U CUBESAT CONCEPT FOR VISIBLE-LIGHT EARTH IMAGING FROM 500 KM LEO

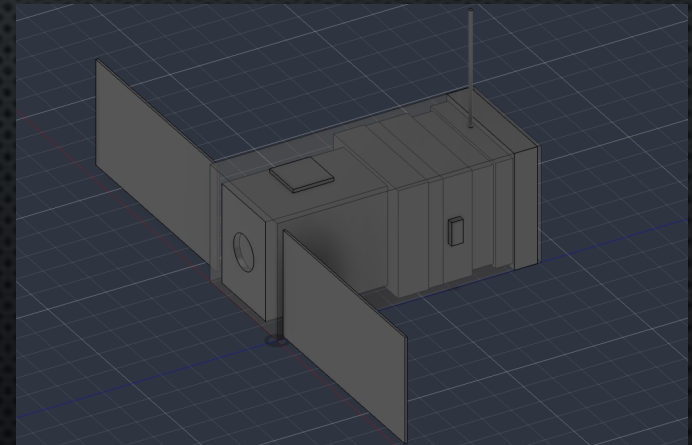


DESIGN EVOLUTION

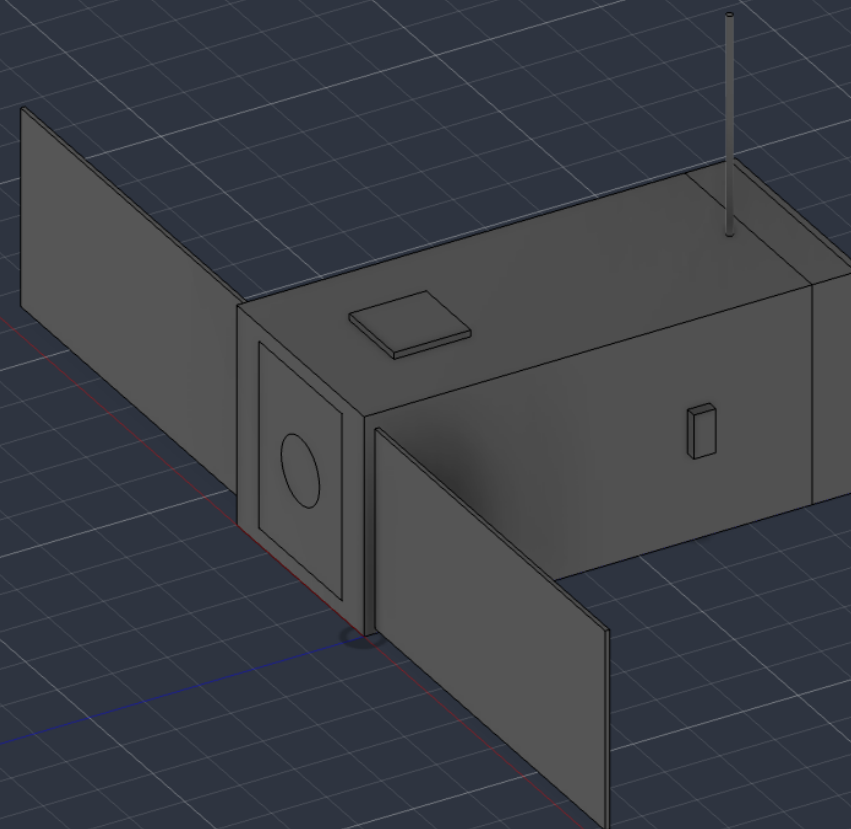
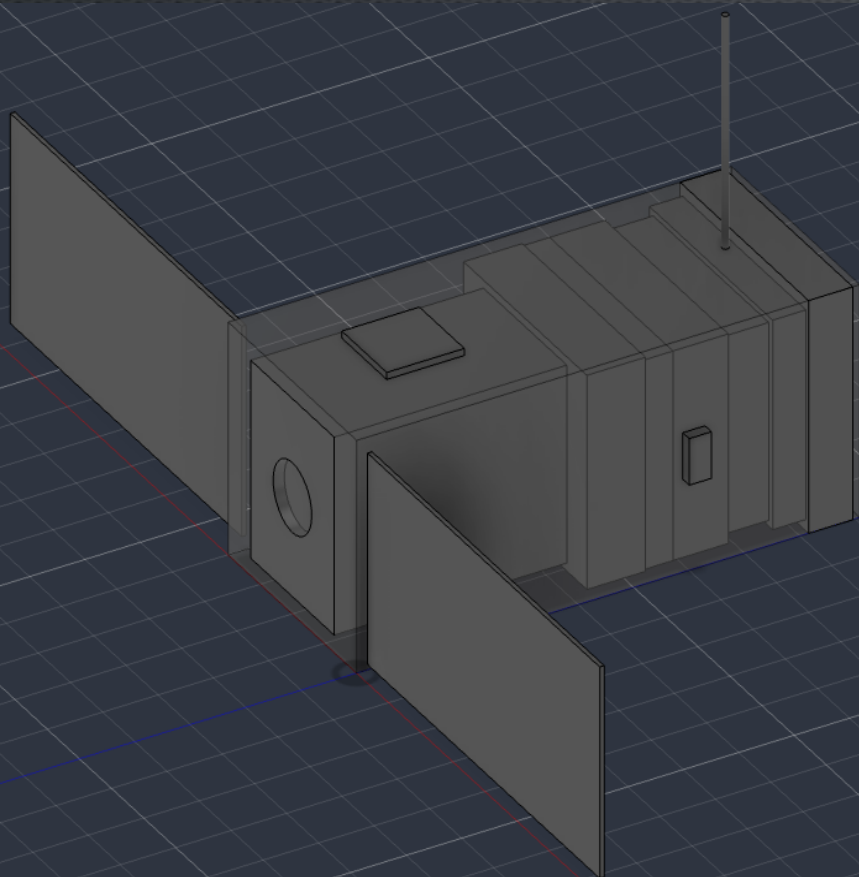
- STARTED AS 1U
- TRADE STUDY SHOWED 2U HAD BETTER MASS/VOLUME/POWER MARGIN



1U: 2.65/5 → 2U: 4.55/5



FINAL ARCHITECTURE



KEY BUDGETS

- MASS
- POWER
- SOLAR TARGET
- ECLIPSE ENERGY

Subsystem	Candidate Hardware	Mass (g)	Power (W)	Status	Notes
Structure	2U chassis	390	—	Preliminary	Commercial 2U reference
EPS	ISISPACE compact EPS Type A	184	TBD	Preferred	Integrated battery/EPS
C&DH	ISISPACE iOBC	94	0.4	Preferred	Flight computer
ADCS	CubeSpace CubeADCS integrated architecture	575 g	1.2 W preliminary avg	Candidate — viable by mass	Controller itself ~0.22 W avg; total system TBD from actuators/modes
UHF Antenna	Deployable antenna	90	—	Preliminary	External
S-band Comms	High-rate transmitter	120	13 peak	Preferred	Image downlink
Payload	Dragonfly Gecko Imager	500 g max design allowance	TBD	Preferred	RGB Earth observation; ~39 m GSD @ 500 km; 80 km swath
Thermal	Thermal materials	40	—	Estimate	
		918			

Subsystem	Mode	Estimated Power (W)	Duty Cycle	Average Power (W)
C&DH	Nominal	0.4	1	0.4
EPS	Nominal	0.3	1	0.3
ADCS	Nominal	1.2	0.75	0.9
Communications	Stand by	0.5	0.9	0.45
Communications	Transmit	4	0.1	0.4
Payload	Stand by	0.2	0.95	0.19
Payload	Imaging	2.5	0.05	0.125
Communications	S-band image downlink	11	0.05	0.55

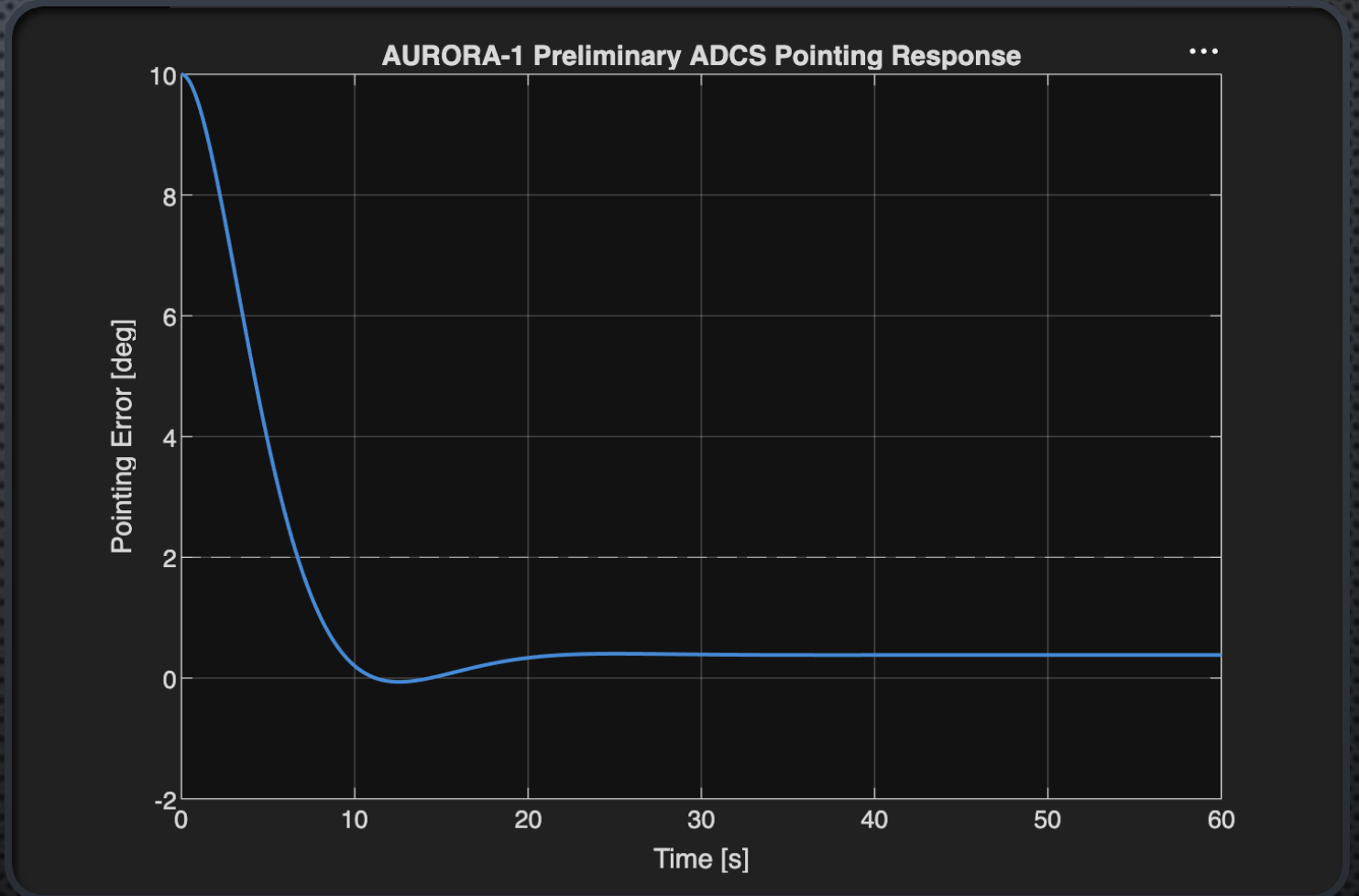
COMMUNICATIONS

- UHF FOR COMMAND/TELEMETRY
- S-BAND FOR IMAGE DOWNLINK
- 100 KBPS MINIMUM, LINK MARGIN RESULTS



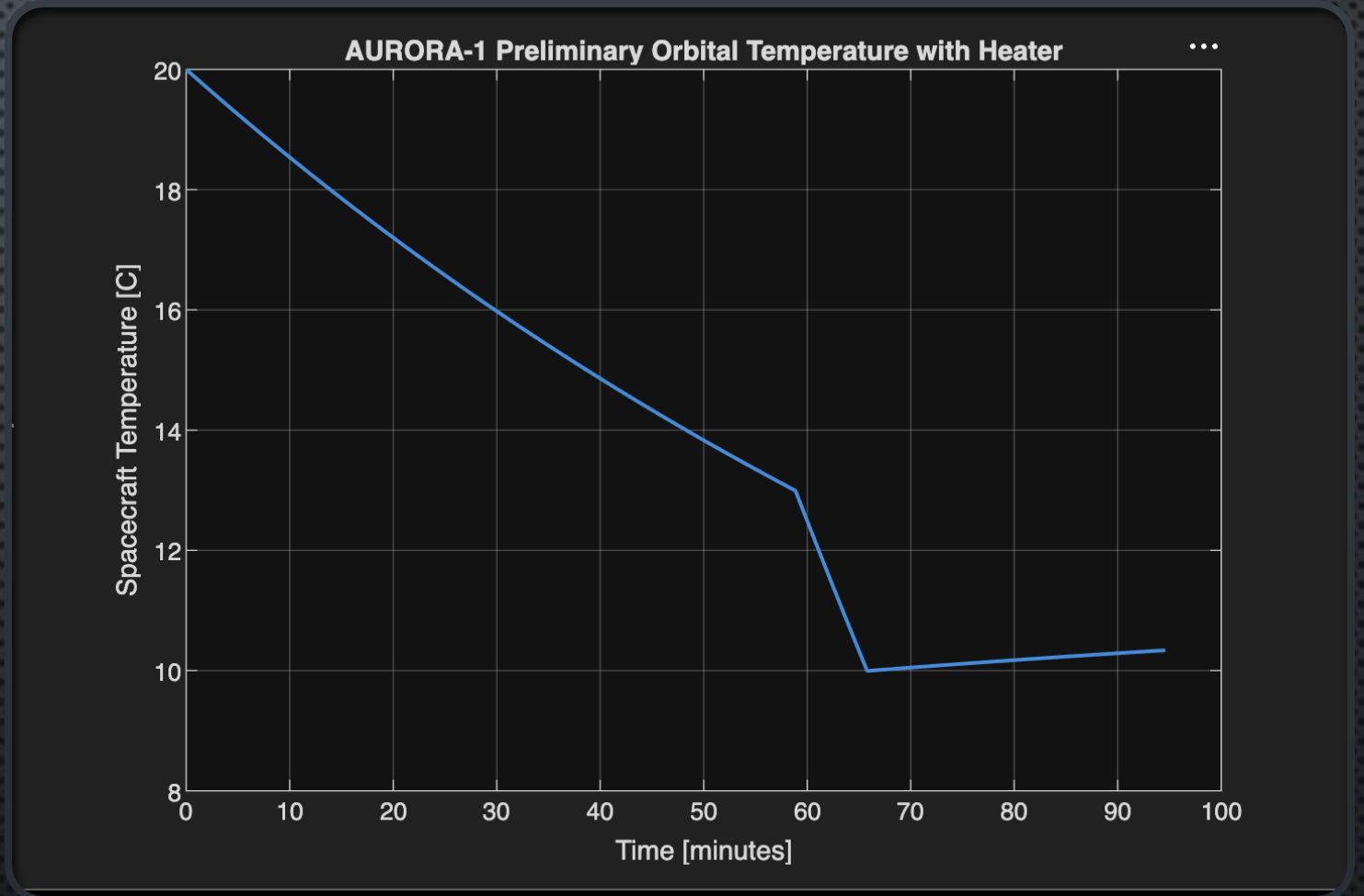
ADCS

- $\pm 2^\circ$ POINTING REQUIREMENT
- 90° SLEW: 6.22 s X/Y, 3.54 s Z
- FINAL POINTING ERROR: 0.382°

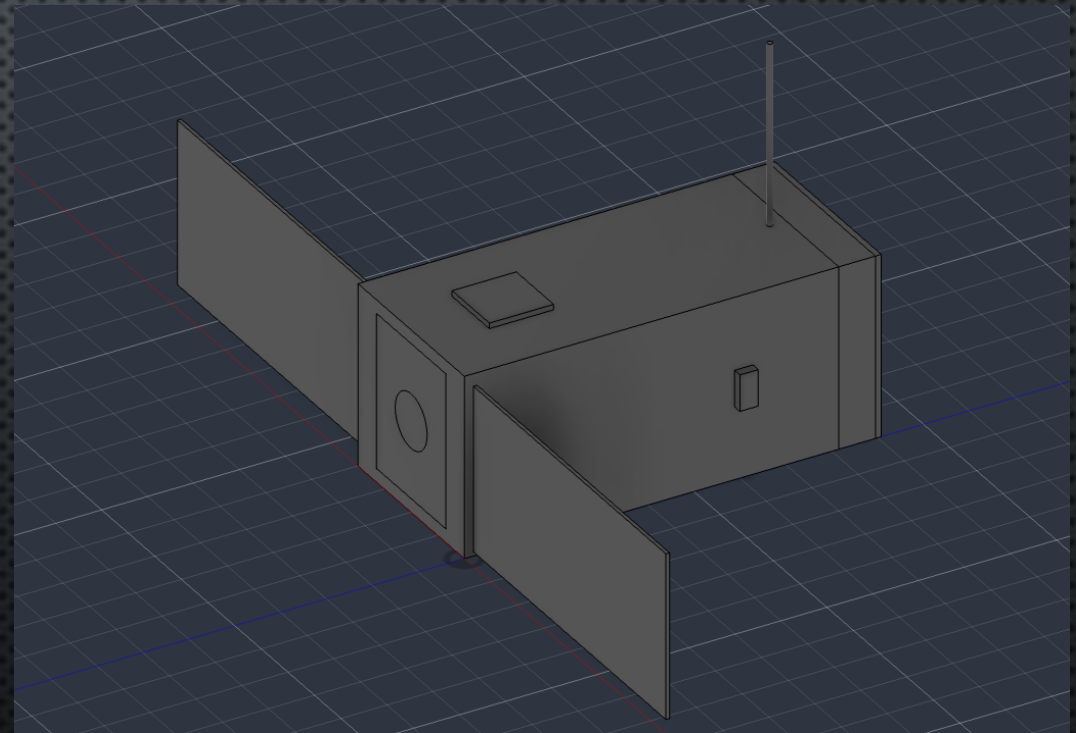
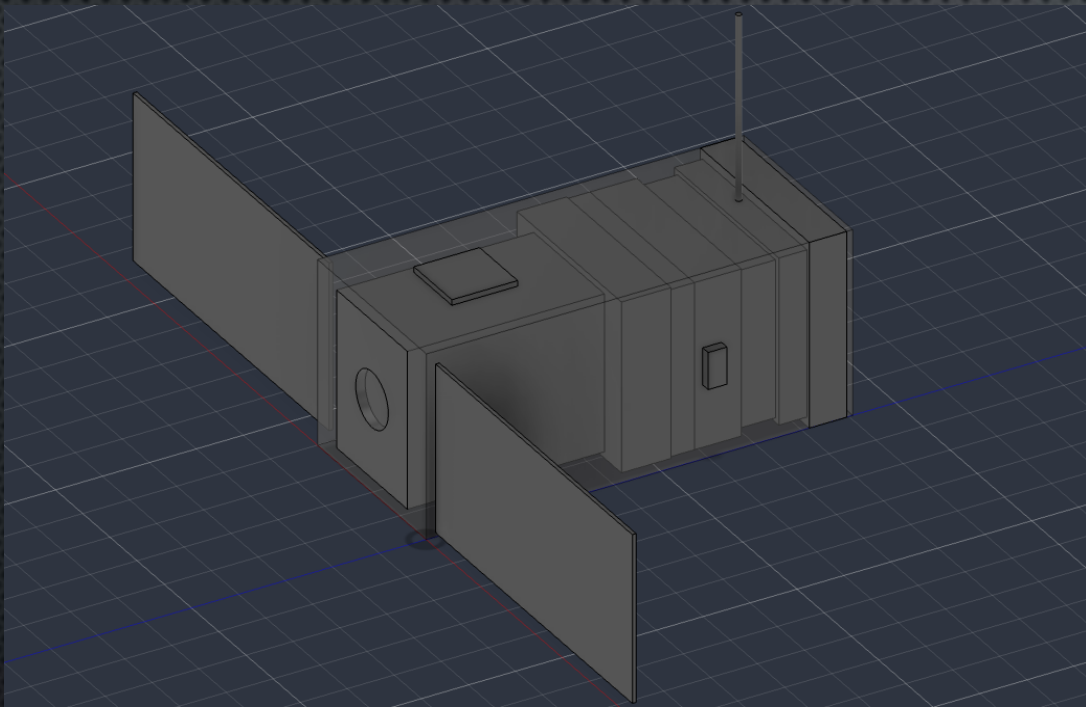


THERMAL

- TRANSIENT MODEL
- EARTH IR/ALBEDO ADDED
- HEATER MODEL HOLDS NEAR 10°C



FINAL CONFIGURATION



FUTURE WORK

